

Beyond Inflammation: Investigating the Immunomodulatory Effects of Metformin in Autoimmune Disorders

Abstract: Metformin, a biguanide commonly prescribed for Type 2 Diabetes Mellitus, is primarily known for its glucose-lowering effects via activation of AMPK and suppression of hepatic gluconeogenesis. However, a growing body of evidence suggests its pleiotropic effects extend to various other physiological processes, including anti-cancer and anti-aging properties. This paper reviews emerging data on Metformin's direct and indirect **immunomodulatory effects**, particularly its potential to influence pro-inflammatory cytokine production and T-cell differentiation. We highlight recent findings from both in-vitro and animal models of autoimmune diseases (e.g., Rheumatoid Arthritis, Systemic Lupus Erythematosus) where Metformin has shown a capacity to reduce disease severity independent of glucose regulation. This evidence strongly supports a hypothesis for **repurposing Metformin** as an adjunctive or primary therapeutic agent in certain autoimmune disorders, leveraging its unique immunometabolic actions.

Introduction: Metformin's role in diabetes management is well-established. Yet, its full therapeutic potential remains largely untapped. Recent research has illuminated its broader effects on cellular metabolism and inflammation, sparking interest in its application beyond endocrinology.

Metformin and Immune Cells: Studies show that Metformin can directly impact immune cell function. It modulates macrophage polarization towards an anti-inflammatory M2 phenotype and can suppress the proliferation and differentiation of pathogenic Th17 cells, which are crucial drivers of many autoimmune conditions. It also influences the metabolic programming of T-cells, shifting them towards a less inflammatory state. These effects are thought to be mediated partly through AMPK activation, which senses cellular energy status and orchestrates metabolic adaptations.

Evidence in Autoimmune Models: In animal models of Rheumatoid Arthritis, Metformin treatment has been shown to reduce joint inflammation and bone erosion. Similarly, in murine models of Systemic Lupus Erythematosus (SLE), Metformin lessened kidney damage and autoantibody production. These effects were observed even in non-diabetic models, suggesting a direct immunomodulatory mechanism. Key inflammatory markers like TNF- α and IL-6 were consistently reduced.

Hypothesis for Repurposing: Based on its demonstrated ability to modulate immune cell metabolism and reduce inflammation in autoimmune models, we hypothesize that

Metformin could be repurposed for autoimmune diseases, particularly those characterized by chronic inflammation and altered immune cell metabolism. This represents a significant shift from its traditional endocrine focus and opens avenues for new combination therapies or monotherapies where existing treatments have limitations.

Conclusion: Metformin's immunomodulatory properties are a compelling area of research. Its established safety profile and widespread availability make it an attractive candidate for repurposing in autoimmune conditions, potentially offering a novel, metabolically-targeted therapeutic strategy.